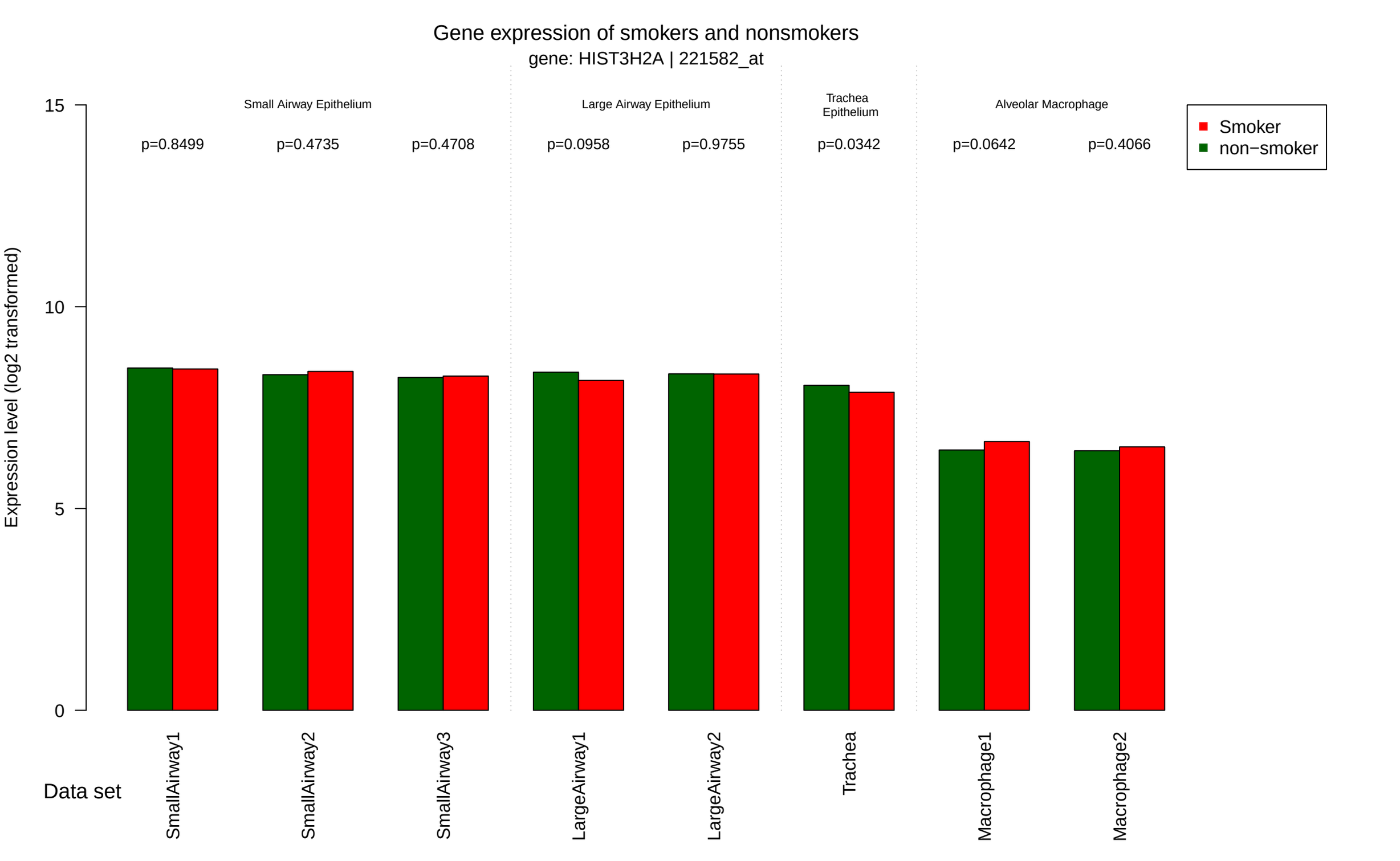
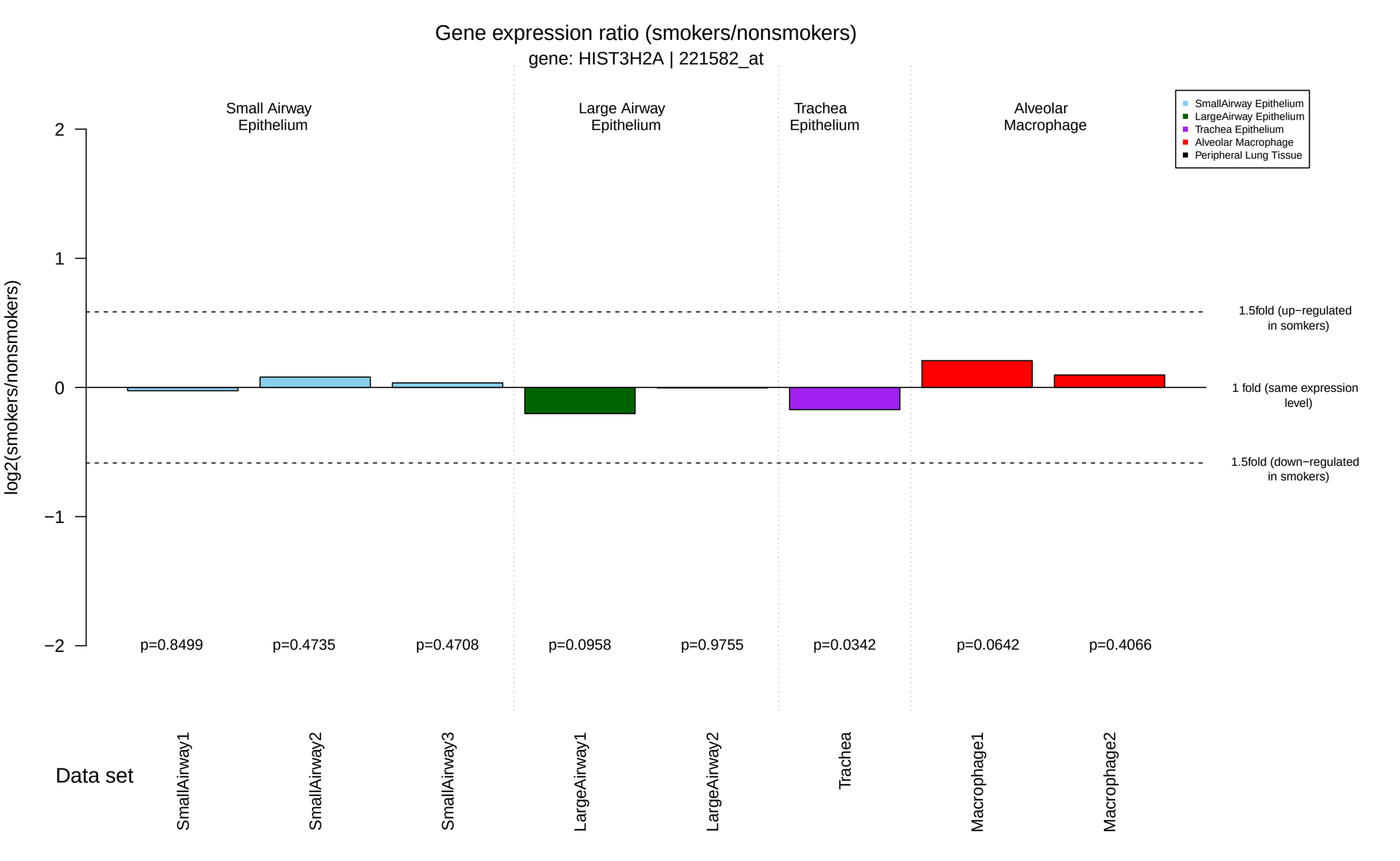
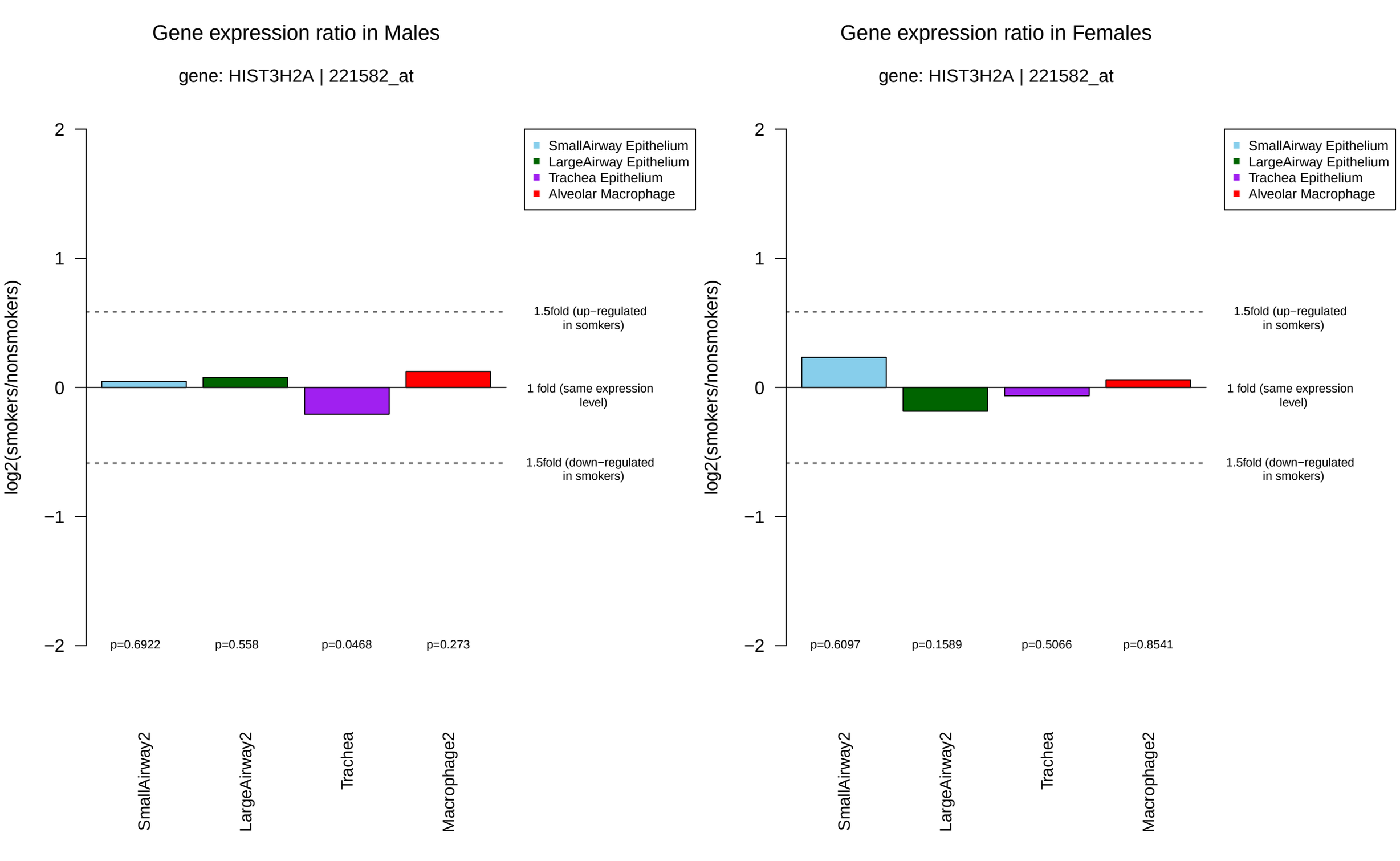
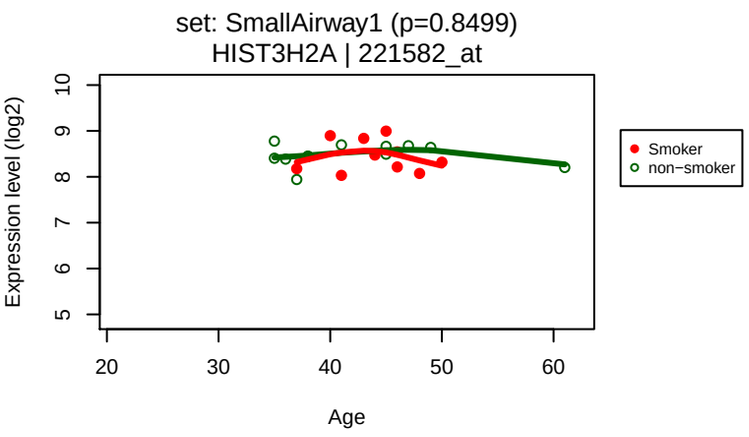
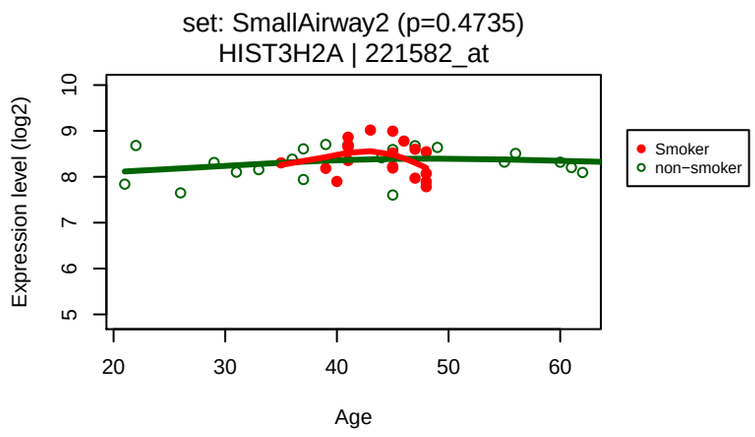
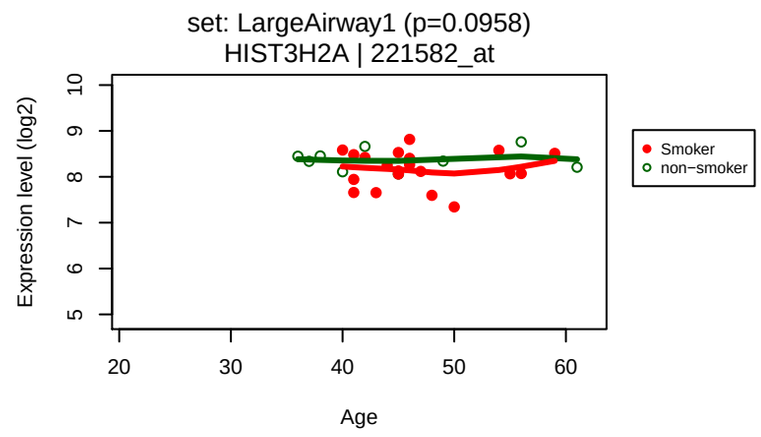

Gene expression of smokers and nonsmokers
gene: HIST3H2A | 221582_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.8499
p=0.4735
p=0.4708
p=0.0958
p=0.9755
p=0.0342
p=0.0642
p=0.4066
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

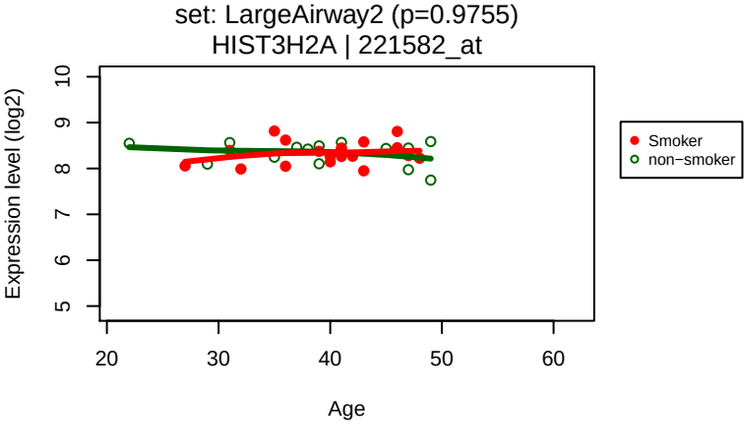

Gene expression ratio (smokers/nonsmokers)
gene: HIST3H2A | 221582_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8499
p=0.4735
p=0.4708
p=0.0958
p=0.9755
p=0.0342
p=0.0642
p=0.4066
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

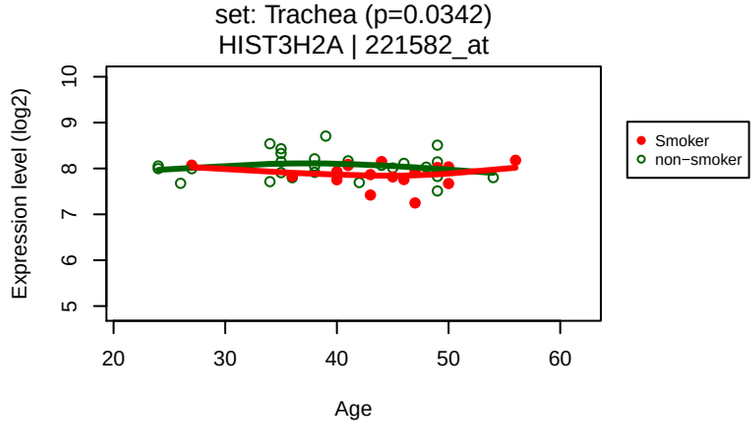

Gene expression ratio in Males
gene: HIST3H2A | 221582_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6922
p=0.558
p=0.0468
p=0.273
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HIST3H2A | 221582_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6097
p=0.1589
p=0.5066
p=0.8541
SmallAirway2
LargeAirway2
Trachea
Macrophage2

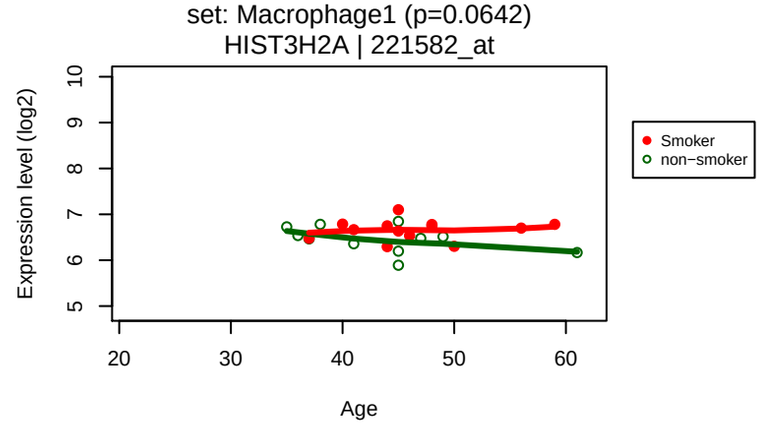

set: SmallAirway1 (p=0.8499)
HIST3H2A | 221582_at
Expression level (log2)
Age
Smoker
non−smoker

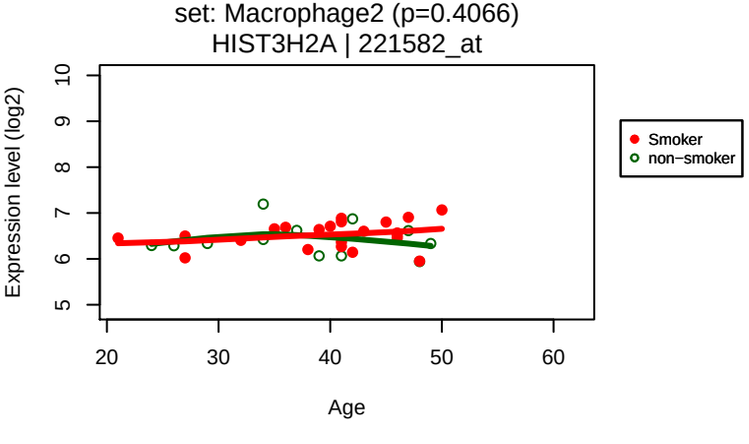

set: SmallAirway2 (p=0.4735)
HIST3H2A | 221582_at
Expression level (log2)
Age
Smoker
non−smoker

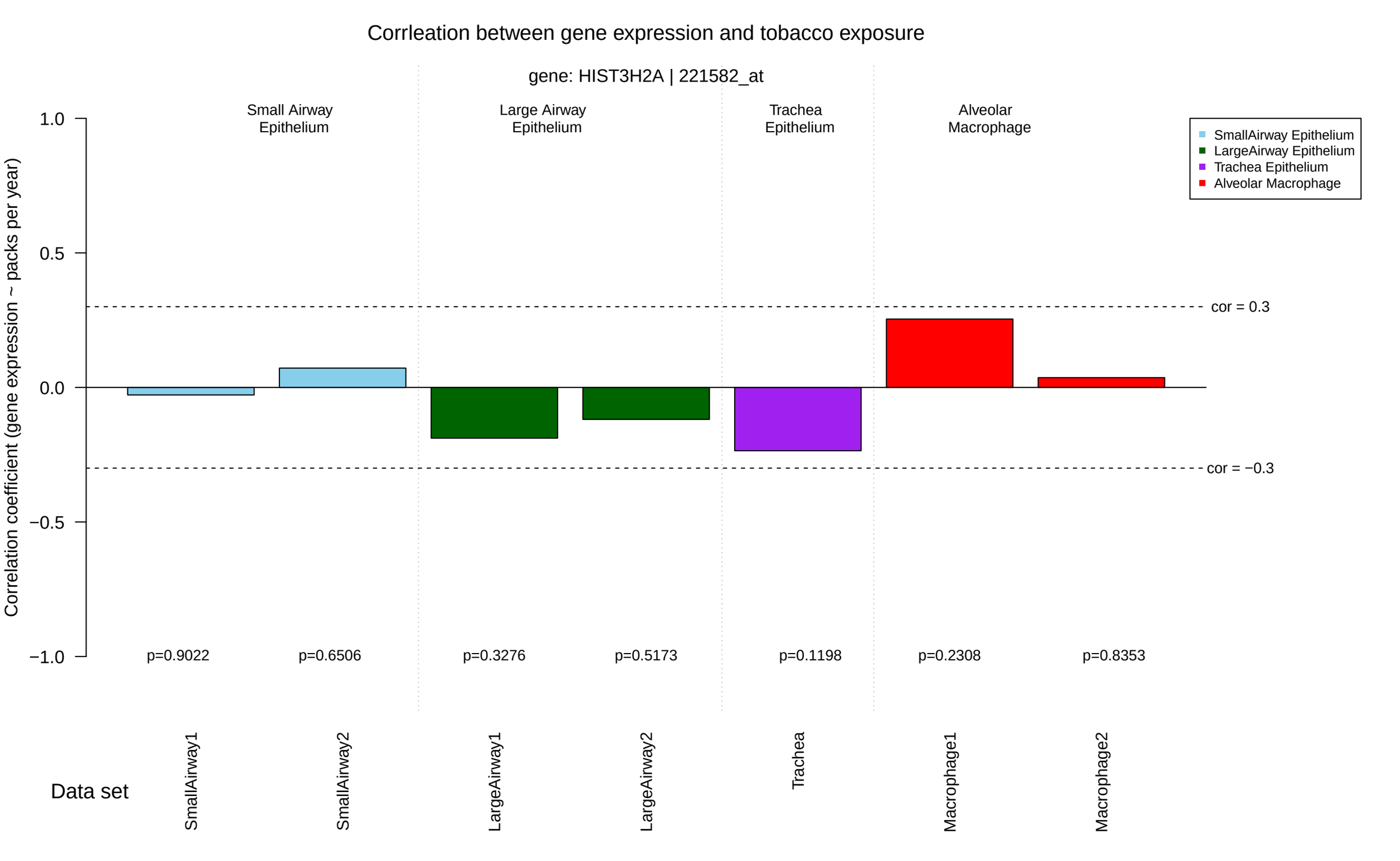

set: LargeAirway1 (p=0.0958)
HIST3H2A | 221582_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9755)
HIST3H2A | 221582_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0342)
HIST3H2A | 221582_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0642)
HIST3H2A | 221582_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4066)
HIST3H2A | 221582_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HIST3H2A | 221582_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.9022
p=0.6506
p=0.3276
p=0.5173
p=0.1198
p=0.2308
p=0.8353
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2